Code No: 115DT **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
COMPUTER NETWORKS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	List the layers in TCP/IP network model.	[2]
b)	Explain guided transmission media.	[3]
c)	What is a router?	[2]
d)	What are collision free protocols?	[3]
e)	What is static routing?	[2]
f)	Explain the reasons for occurrence of congestion in networks.	[3]
g)	What is Address Resolution Protocol (ARP)?	[2]
h)	What is CIDR?	[3]
i)	Explain DNS.	[2]
j)	What are services provided by application layer?	[3]

PART - B

(50 Marks)

2.a) What are the reasons for using layered protocols?
b) What is OSI model? Explain in detail the functions of different layers in OSI model. [5+5]

OR

3.a) Explain the operation of sliding window protocol.
b) Explain the design issues in data link layer. [5+5]

4.a) Explain CSMA with collision detection.
b) Briefly explain the operation of ALOHA system. [5+5]

OR

5.a) Describe the Ethernet standards.
b) Explain the purpose of spanning tree bridges in mac sub layer. [5+5]

6.a) Using an example network, explain distance vector routing.
b) Explain congestion control mechanisms at network layer. [5+5]

OR

7.a) Enumerate the difference between virtual circuit and data gram subnets.
b) Explain hierarchical routing. [5+5]

8.a) In detail, explain the different addressing schemes in computer networks.
b) Explain features and purpose of ICMP protocol. [5+5]

OR

9.a) Explain crash recovery in transport layer.
b) Briefly explain the operation of DHCP. [5+5]

10. Explain the TCP connection establishment, TCP connection release, and TCP connection management. [10]

OR

11. With respect to application layer, write short notes on:
a) Client server model
b) SSH [5+5]

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